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Reconsidering perceptual constancy

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ABSTRACT

The world shows some degree of invariance, and we perceive this invariance despite a lot of variation generated locally by our movements, changes in illumination, and the way in which our sense organs react to stimulation. Generally, philosophy and psychology each explain our perception of invariance through the notion of ‘perceptual constancy’. According to the traditional definition, perceptual constancies are capacities to perceive the objective (i.e., perceiver- and context-independent) local properties of external objects despite variation in the stimulation of sensory organs. In this paper, we argue that the traditional understanding of perceptual constancy should be expanded to include a broader range of phenomena connected to the perception of invariance. Our argument starts from an odd activity: pole balancing. Balancing a pole-shaped object (e.g., a broomstick) on the palm of your hand in order to maintain it in an upright position requires (multimodal) perception and continuous tracking (within a certain range) of a particular kind of global invariant feature in the world. By examining this activity and its connection to the notion of perceptual constancy as it is traditionally understood, we propose an alternative account with the goal of clarifying the relationship between the alleged mechanisms underlying constancy and the philosophical implications of the phenomenon.

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1. Introduction

When we perceive the world, our sensory organs get stimulated in unstable and inconsistent ways, due to the constant flux of energy being exchanged between our bodies and the environment. Time passes; sunlight comes and goes; we move our bodies at different speeds and in different directions, occupying many different locations in space, and modifying the locations of other material objects; our eyes are always moving; our attention shifts, just like our bodily balance, and so forth. Nevertheless, even in such a ‘Heraclitean’ world, we don’t perceive everything as *always*

changing in *every* aspect. Against all odds, we perceive many features of our environment as remaining invariant, even as change, sometimes caused and controlled by us ‘restless’ perceivers, happens all around them. Even though we will eventually argue that perceived invariance does not only depend on specific fixed features of the world, for now we can talk as if such features are indeed out there, independent of us, and when everything works properly, their stability and invariance is manifest in our perceptual experience. Following the tradition of ecological psychology, let’s call these features “invariants” (e.g., Amazeen & Turvey, 1996; Gibson, 1979; Shockley et al., 2004). Next, let’s suppose that what changes over time due to movement or other environmental changes are also features of some sort; we can call them “variants”. While invariants are distal, that is, experienced as external to us, many variants are proximal, that is, they are experienced as features of us or dependent on us; ways in which our sensory organs stay in contact with the world and make it possible to perceive the invariants in it. For example, we can perceive a table’s shape as an invariant while we move around it while at the same time being aware that the apparent shape causally connected to the image projected on my retina changes. Moreover, it seems that the invariance of the shape is experienced *in virtue of* the way in which the viewpoint-dependent retinal projection changes: namely, consistently with how the relevant spatial relations between us and the table (and, thus, the visual angle) change. Similarly, we can perceive that same table’s color as an invariant while illumination in the room changes, and this will be despite – or rather, in virtue of – the apparent color projected to the retina changing consistently with how the light in the room changes. We only perceive invariants, that is, insofar as our perceptual system is sensitive to and appropriately processes the corresponding variants. Variance and invariance are, in other words, co-defined and interdependent: invariance is only invariance when contrasted with variance, and the presence of invariants is only made possible by the presence and registration of variants.

“Perceptual constancy” is generally the term used in philosophy and psychology to capture the capacity of an animal’s perceptual system to be aware of and respond to distal invariance despite proximal variance in sensory stimuli. But even though the notion of invariance itself is quite broad, the definition of perceptual constancy accepted by most psychologists (and those philosophers relying on their theories) does not seem to be as broad. Indeed, perceptual constancy is usually understood as the capacity to represent specific local properties of external objects (e.g., colors or shapes) as they are independently of the context and the perceiver, and to do so by ‘overcoming’ and ‘correcting’ the variation in the stimulation of

sensory organs (Hochberg, 1988; Palmer, 1999; Rock, 1975). In turn, the invariance perceived at the personal level consists in those represented local properties of objects.

This paper aims to put this idea under scrutiny and to suggest an alternative to it. In particular, we will structure our arguments around three questions: Is the perception of stable object properties the only or even the main phenomenon a theory of perceptual constancy should be concerned with? Is *all* the invariance we experience associated with local properties (e.g., shape or color) of specific objects? And how should we interpret the idea of perceiver-dependent proximal variation being ‘corrected’ or ‘compensated for’ in light of the interdependence of variance and invariance? By putting pressure on the traditional understanding of perceptual constancy as a way to account for perceived invariance, we hope to convince philosophers and psychologists interested in the issue to reconsider the traditional theory and to make the concept of perceptual constancy more explanatorily powerful by applying it to a broader group of perceptual invariance phenomena.

2. The traditional view

In the past decade or so, a handful of philosophers have talked about perceptual constancy and the role it plays in perception more generally.¹ The very definition of this capacity, however, is rarely the main discussion topic. Jonathan Cohen (2008, 2015) claims that “the textbook characterization has it that perceptual constancy is nothing more or less than stability in perceptual response across a range of varying perceptual conditions” (2015, 623).² As a matter of fact, this definition is not very different from what we started the paper with: it states that constancy allows an animal to respond stably to features of the world (where ‘response’ here means ‘form a perceptual representation’) while the conditions in which the features are presented (i.e., the intrinsic properties of proximal sensory stimuli) change. However, psychologists have often interpreted the textbook definition more narrowly. For example, psychologist Irvin Rock (1975) described constancy thus: “Perceived qualities such as color, size, and the like tend to remain constant despite the fact that the proximal stimulus, for example, the retinal image of objects, is continually changing” (as cited in Hatfield, 2009, p. 181). Another psychologist, Hochberg (1988; as cited in Brown, 2003) defines perceptual constancy as “the constancy of perception of the fixed properties of distal objects, despite variation in the proximal stimuli from objects.” (251–252). Burge (2010, p. 408) defines constancy somewhat similarly: “Perceptual constancies are capacities systematically to represent a particular or an attribute as the same despite significant variations in registration of proximal stimulation”. All these definitions express

a similar thought: perceptual constancies are capacities to represent local features of particulars (attributes) as invariant *despite* variation in the proximal stimuli caused by such features.

A second – and related – important commitment of the traditional view is that perceptual constancy mechanisms perform certain kinds of inference-like operations which have proximal stimuli as inputs and (possibly accurate, i.e., objective) representations of local properties of particulars as outputs. Indeed, much of the scientific research on perception seems to assume that the constancies have the goal of solving so-called *underdetermination problems*. Underdetermination problems arise because – assuming that the task of the perceptual system is to represent accurately and objectively how the world is, i.e., which mind-independent objects and properties are out there – not enough information about such objects and properties is immediately available through sensory stimulation. Proximal stimuli are ambiguous: information uniquely specifying perceiver-independent properties of objects is generally ‘mixed up’ with information about other features of the perceptual scene (i.e., the context, including the perceiver herself). Many different combinations of objects, properties, and contexts could have caused the same proximal stimulus at any given moment. For instance, various trapezoidal tables and a rectangular one can all cause the same two-dimensional retinal image when viewed from certain angles, or many different combinations of light intensity and surface spectral reflectance properties can cause the same color appearance. Hence the idea that the proximal stimulus *underdetermines* the distal properties of objects, making perception a matter of resolving such underdetermination problems for the sake of achieving accurate objective representations, with constancy mechanisms being the ones designated for the job. According to Rescorla (2015), indeed, “How does the perceptual system achieve constancies? By using “implicit assumptions” to *discount* variations in proximal stimulation” (2; italics ours).

According to the traditional view, then, perceptual systems solve underdetermination problems by decomposing the proximal stimulus into (hypothesized) simpler components. This allows the system to subsequently isolate the local property to be attributed to the object in question. The ‘irrelevant’ components of the proximal stimulus are thus first *estimated*, then *discounted*, so that the relevant property can be represented. The traditional view, thus, combines two claims:

- (1) The function of perceptual constancy mechanisms (and thus the very definition of perceptual constancy) consists in achieving accurate objective representation of local properties of particulars.
- (2) The process through which this function is performed is one of “reverse inference” aimed at resolving proximal underdetermination.

Our dissatisfaction toward the current version of the traditional picture is rooted in the following observation: there seems to be an intuitive asymmetry between the broad, ‘commonsense’ notion of invariance we started with and the ‘technical’, narrower, and more fine-grained notion of perceptual constancy found in theoretical accounts. With some notable exceptions (e.g., Buccella, 2021; Chemero, 2009;; Gibson, 1979; Ittelson, 1951; Kilpatrick, 1952; Noble, 1981), so far this asymmetry hasn’t been sufficiently thought through (or even fully acknowledged, in fact), and this has translated into problematic implicit assumptions within mainstream philosophy of perception. Mainstream philosophers of perception do not seem to fully acknowledge the important role temporally extended, sometimes actively generated proximal variance plays in enabling the perception of the external world, and do not fully appreciate the many different types of invariant features that figure in ordinary perceptual experience and contribute to its phenomenal character.

In what follows, we discuss an odd but surprisingly complex and psychologically interesting activity: pole balancing. Balancing a pole-shaped object (e.g., a broomstick) on the palm of your hand in order to maintain it in an upright position requires a quite sophisticated combination of visual and haptic perception, and scientists have demonstrated that subjects are capable of pole balancing in virtue of their (multimodal) perception and continuous tracking (within a certain range) of a particular kind of ‘global’ invariant feature across variation in tactile and visual proximal stimulation over time.

We argue that pole balancing involves the display of some kind of perceptual constancy capacity in virtue of the role played by experienced invariance, and yet that the current version of the traditional view would not accept this verdict due to their narrow and overly specific definition of perceptual constancy. Indeed, the invariant feature perceived during pole balancing – and on which successful pole balancing directly depends – does not seem to correspond to any local objective property of any of the objects involved in the activity. Rather, the experienced invariance seems to be tied to a global relational feature of the ‘pole + agent + context system’, whose invariance (within a certain range) in this particular circumstance is both controlled and experienced by the balancer through her visual system coordinating with other sensory systems (specifically, haptic, proprioceptive, and vestibular) in creating regular patterns of proximal variance. If perceptual constancy is supposed to be the capacity that enables and explains perceived invariance, then the mechanisms responsible for the tracking of global relational invariants like the one found in pole balancing should count as mechanisms of perceptual constancy, and this in turn implies that the traditional view of constancy must be revised, since, in its current version, it is (a) too focused on vision and on an ‘attribution-of-

properties-to-substances' conception of perception, and (b) neglecting the temporal dimension of invariance perception and the consequent positive role of proximal variance in enabling it.

3. Pole balancing and invariance perception

Consider the admittedly unusual activity of pole balancing. In pole balancing, one balances a broomstick upright on one's hand. The pole is essentially an inverted pendulum, with the attachment point at the hand, the low point, and the mass elevated in the broomstick.³ The perfectly upright position for the broomstick is an unstable equilibrium point, so the pole is constantly in motion while at the same time remaining within a certain spatial range.

Clearly, pole balancing requires multimodal – both visual and haptic – perception to guide the balancer's movements in the attempt to keep the pole in position; also clearly, keeping the pole within the (unstable) equilibrium range requires something in the balancer's perceptual experience to be variant and something to be invariant. Moreover, the variant and invariant components of the experience are related in constitutive, lawful ways: the constant motion of the pole + balancer system, and the perceptual variants generated thereby, are precisely what allows the system to be nonetheless in a state of equilibrium, i.e., a state in which invariance and stability are actively maintained. But *what* does the experienced invariance consist in, exactly? According to Foo et al. (2000), pole balancers directly perceive the variable 'time-to-balance', i.e., the ratio of the angular distance between being perfectly upright and the position of the pole at some time (q) to the rate of change of that angular distance (dq/dt). Successful pole balancing requires experiencing $d(\text{time-to-balance})/dt$, that is, a *global relational property* of the coupled system composed by the balancer and the pole, as being relatively constant (i.e., within a critical range) while the values of the individual quantities (q , dq/dt) are in constant flux (hence, the equilibrium point being unstable).

At the beginning, we suggested that our stable perceptual experiences depend on our perceptual system's capacity to track invariance in the environment and to respond to it while proximal sensory stimuli exhibit a lot of variance. We also suggested that the notion of perceptual constancy is traditionally understood, while aiming at explaining how invariance is perceived and what this invariance consists in, is not equipped to account for *all* kinds of invariant features that contribute to stable perceptual experiences. Indeed, on the one hand, activities like pole balancing require the perceptual system of the balancer to be sensitive to some kind of (dynamic) invariant feature in the environment, while, on the other hand, such an invariant feature does not fit the traditional definition of perceptual

constancy, since it is not a stable property of a specific object, but rather a global relational-structural feature of the balancer + pole coupled system. Based on how psychologists usually account for the perceptual capacities involved in an activity like pole balancing, it seems clear to us that constancy mechanisms should be part of the explanation. However, if this is the case, constancy mechanisms would have to look functionally quite different from how the traditional view describes them.

In addition to the fact that the invariant feature relevant to explain pole balancing is a relational feature of a system rather than a stable property of a single object, notice that this global relational feature stays *roughly* invariant, but never exactly so. Rather, as long as the pole is in balance, this feature's value varies while remaining within a certain *range*, which indicates that representing the feature as having specific values for every variable involved would not explain or enable successful pole balancing. One possible solution would be to claim that perceptual systems represent ranges instead of – or in addition to – determinate magnitudes,⁴ but this would then raise questions like: Are the boundaries of the range represented fixed? How are they determined? And if the goal is to solve underdetermination in proximal stimuli, how does representing ranges, indeterminate entities *par excellence*, help? It is possible that the tradition is able to accommodate these concerns, but so far, they hadn't even ever been raised.

Moreover, the invariant feature tracked during pole balancing is not related to proximal sensory stimulation in the way required for an underdetermination problem to arise. Proximal stimulation in the case of pole balancing might well be ambiguous with respect to its external causes, but such ambiguity is irrelevant, because the source of the experienced invariance isn't, strictly speaking, the proximal stimulus' external cause. All these observations are in tension with the traditional view. Therefore, mainstream philosophers of perception interested in constancy and invariance perception have two options to choose between: either they maintain that the invariance experienced during pole balancing has nothing to do with constancy mechanisms – a reply that, based on how we have intuitively characterized invariance perception, would require additional justification in order not to seem ad hoc – or they acknowledge that the traditional definition is explanatorily inadequate.

Going back to pole balancing, notice that balance relies on both variance and invariance: because of the nature of the inverted pendulum structure, the pole can only be maintained in balance if the balancer and, consequently, the pole itself, are in constant movement. A feature of the world that is in balance is a feature that shows at least some degree of invariance against a background of variance, and whenever balance is experienced, invariance is, too.

Moreover, as we already pointed out, variant and invariant features are related in certain law-like ways (and must be experienced that way if balance is being experienced). In pole balancing, the relation between variant and invariant features is particularly evident: the invariant feature *just is* the regularity with which variant features change over time. This takes us to the second reason why we think the traditional conception of perceptual constancy is explanatorily inadequate: if perceptual constancy mechanisms are those enabling the perception of invariance, then it cannot be correct to say that their goal is to estimate and then discount variant features in order to obtain a representation of the invariant feature as a result. This latter claim is exactly what is implied by the traditional account of constancy, which is rooted in the idea that constancy mechanisms fundamentally function to solve underdetermination problems and to recover the objective local properties of particulars. In the following, we explain why we think that there is no principled reason to stick to an underdetermination problem-centered conception of constancy mechanisms instead of re-thinking the very conditions in which invariance is experienced.

Our argument starts from the main motivation for a traditional account of constancy mechanisms, that is, the fact that we have the capacity to issue accurate perceptual judgments about the objective local properties of specific particulars (e.g., the size of an object at varying distances). Indeed, proponents of the traditional account have assumed (based on experiments in perceptual psychology) that perceptual judgments (reports) constitute evidence of what is represented in the subject's own perceptual experience, so that if, say, a subject attributes the correct size to an object after seeing it at various distances, this must be because the subject's visual system has successfully managed to represent the object's objective size. However, there might be a different explanation for our ability to issue accurate judgments about local properties of specific particulars, which does not rely on the claim that constancy mechanisms themselves work by representing those properties.

Most non-philosophers (as well as many philosophers and psychologists) would probably take as a paradigmatic example of a successful display of perceptual constancy something like the image below. In the picture, you can 'see' that all the trees have more or less the same size, even though they all occupy differently sized portions of the visual field. You 'see' that they are all of the same size, presumably, because your visual system is successfully displaying size constancy. The traditional view would say that successfully displaying size constancy amounts to the visual system recovering from the retinal image a representation of the perceiver-independent and

perspective-independent size shared by all the trees. Once this representation is available, the visual system ‘attributes’ it to all the trees, after having discounted information about distance.

Despite the intuitive plausibility of this case, it is a mistake to consider it a good starting point to understand what size constancy is and how it works. Indeed, our judgments about perceiver-independent properties of objects such as size when facing an image like [Figure 1](#) might actually be knowledge-based and inferential, rather than purely based on perceptual phenomenology (which in turn, according to most philosophers of perception, is entirely fixed by representational content).

For Granrud ([2004](#), [2009](#), [2012](#), [2006](#)) showed that in situations analogous to [Figure 1](#), adults issue more accurate ‘objective’ size judgments than children. He argues that during development we learn to interpret phenomenal experience through the lens of what we *know* about how objects typically look when they are far away, namely, that they appear smaller even though they have the same size as when seen up close. In viewing images like [Figure 1](#), it is plausible that, knowing the effects of distance on size appearance, ‘mature’ observers who are trying to issue an accurate size judgment will rely on explicit reasoning. They can compare the different



Figure 1. Real-life example of ‘static’ size constancy: the trees, though placed at different distances from the perceiver, still look roughly of the same size.

perspectival appearances of the trees, which are present simultaneously on the retina, and use their knowledge about distance to make the appropriate 'calculations' and arrive at a perspective-independent size estimate. This suggests that the experiences we have in these static, synchronic cases may not result directly from successful displays of perceptual constancy, but from a cognitive inference.

If Granrud is right and, at least in the case of size constancy, we in fact *learn* to cognitively compensate for distance when estimating sizes, the next question becomes: How is this ability acquired? The answer we find most plausible is: by being frequently exposed to cases in which objects move away from or closer to us, without looking as if they are changing size. That is, from infancy, children repeatedly experience moving objects that they are holding in their hands toward and away from their eyes, without having a tactile or visual experience of the objects growing and shrinking. Indeed, a central feature of visual experience is that it is continuous and extended in time, and as time passes the visual scene undergoes changes, in particular *spatial* changes: objects change their position in space, we change our position in space, etc., and we perceptually keep track of these changes through variation in proximal stimuli. Therefore, we think that the strategy leading to accurate size judgments in *static, synchronic* cases like [Figure 1](#) can be learned only by experiencing the invariance of objects' sizes *while* distance changes, that is, in cases that are *dynamic* and *diachronic*.

Accordingly, perceptual constancy is more likely the capacity to latch onto and respond to distal invariants *as proximal stimuli change over time*, while the static cases, on which most scientific research on, for example, color constancy, relies, should not be considered

cases in which the constancies are directly at work.⁵ At most, what happens in static cases fundamentally depends on perceptual constancy being displayed in dynamic cases, in which proximal stimuli change and yet we experience something as remaining invariant in the complex network of relations connecting us to the environment.

It is also worth noting that Grandrud's studies used impoverished, artificial stimuli when asking subjects to estimate sizes. The fact that subjects seemed to rely on reasoning and cognitive inference when attributing sizes *in such conditions* is particularly telling in two ways. First, it suggests that the constancies as *perceptual* capacities do not seem to be easily assessable in the context of an empirical framework which focuses on the attribution of perceiver-independent properties to particulars. Second, it is compatible with our proposal, according to which perceptual constancies are the capacities engaged when

maintaining balance and coordination over time in the way a perceiver relates to her environment, which in turn requires full-cue conditions and ecologically realistic stimuli.

We suspect that proponents of the traditional view would agree with the idea that dynamic constancy is the fundamental notion, though they would probably also hold on to a more central role for static cases. We, however, claim that the capacity of perceptual constancy is fundamentally displayed over time, and that *for this very reason* it is implausible that it itself consists primarily in the recovery of perceiver-independent properties through inference-like operations. The example of pole balancing highlights precisely this point: in order to explain the capacity to perceptually track distal invariance across and during proximal variation, perceptual constancy does not need to consist exclusively and even primarily in the recovery of local objective properties of particulars. Thus, if our suggestions are plausible, the traditional conception of perceptual constancy should be abandoned in favor of broader and more inclusive one.

4. Further remarks

A natural response to what we argued for in the previous section might be that pole balancing is such a strange activity that we should not draw broad conclusions from it. We agree that people don't go around balancing broomsticks on their hands very often, and some people might never even try it. Fair enough: but one of our most common activities is a lot like pole balancing. The majority of humans spend significant periods of time engaged in standing balance, i.e., standing upright and not falling over. Like a balanced pole, a standing human is an inverted pendulum, with the perfectly upright position as an unstable equilibrium point. As one might expect, standing balance is much more thoroughly studied than pole balancing. (For a review, see, Riley et al., 2011.) An upright human is inherently unstable, and that's why humans constantly engage in postural sway, that is, actively controlled motion around and through the unstable perfectly upright equilibrium point.

As is the case with pole balancing, the movement generates the information that is used to maintain standing balance: invariance emerges through actively generated variance, rather than by perceiving specific perceiver-independent properties of objects in the scene. Postural sway consists primarily of small motions, predominantly forward and backward, of the body's center of mass. These motions create a displacement of the head and eyes relative to the environment, and this displacement produces visual changes. These changes are lawfully related to the sway and are used to control the sway and maintain standing balance (Anderson & Chemero, 2019; Faries & Chemero, 2018; Lee & Lishman, 1975; Riley et al., 2011). The

case of standing balance is parallel to that of pole balancing and has a similarly uncomfortable relationship to the traditional view of constancy. Maintaining upright posture is plainly perceptual, and it clearly involves responding to some kind of perceptual invariant, but it depends neither on solving an underdetermination problem nor on discounting local variation to perceive distal invariant features.

There is a crucial way, however, in which maintaining upright balance differs from pole balancing that is also important for our overall argument. The visual variants generated by postural sway enable the swaying balancer to maintain a relatively invariant distance from surrounding objects (during standing balance, sudden, significant variance in the distance of objects means that you are falling). In fact, it turns out that the visual changes created by postural sway are crucial for perceiving distance. When you are moving, there is no ambiguity about the distance of objects and, crucially, no need to estimate their distance from prior probable knowledge about their expected sizes: the visual changes caused by this small movement make the distance wholly unambiguous and directly perceivable (see, Anderson & Chemero, 2019; Balasubramaniam et al., 2000; Faries & Chemero, 2018).

The key point here is that this is not a matter of successfully estimating an 'objective' perceiver-independent local property (i.e., absolute distance between the object and the perceiver) despite the interference from local variants; rather, the active creation of local variants makes the perception of invariance in the environment possible.

5. Conclusion

As mentioned at the beginning, perceptual constancy is usually the notion used to capture the capacity thanks to which we can perceive what in the world remains (relatively) stable and invariant independently of how a particular perceiver's sensory system in particular circumstances is stimulated. The traditional conception of perceptual constancy is rooted in the notions of underdetermination problem, reverse inference, and localized attribution of properties to particulars. We have argued that the traditional view neglects those instances of invariance perception which do not fit their narrow definition of perceptual constancy: The examples we presented show that much – if not most – of the perceived invariance in the world is not ascribable to the objective representation of local properties of particulars. Rather, perceived invariance includes features that specify the world's relations to the perceiver, which are constantly in flux. With her movements, the perceiver creates the diachronic patterns of proximal variation which specify and grant access to what's invariant within the system she is

part of. If all of this is correct, then the traditional account must be broadened, and a more inclusive definition of perceptual constancy must be formulated.

Notes

1. In addition to Cohen and Burge, we can include in this group Hatfield (2009) and Matthen (2010).
2. Even though Cohen doesn't talk in terms of constancy being a capacity, we can read him as implying that constancy, in his view, is the 'outcome' of what we (following Burge) have defined as a capacity. Sometimes a capacity and the capacity's 'product' are referred to using the same term. For example, the capacity for bike riding 'results' in the action of bike riding, and what we call 'bike riding' is, depending on the context, either the capacity or the actual action. Similarly, we suggest that Cohen's definition refers to the phenomenon in which displaying the capacity for constancy consists in.
3. Compare: the (non-inverted) pendulum in a grandfather clock, which has the attachment point at the top and the mass hanging below.
4. A thesis that, indeed, some philosophers have tried to argue for (e.g., Boone, 2020, 2020; Nanay, 2010; Stazicker, 2011).
5. For empirical work on color constancy done within the traditional framework, see, (Brainard, 1998; Maloney, 1986; Maloney & Wandell, 1986).

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No potential conflict of interest was reported by the author(s).

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Tony Chemero got his Ph.D. in Philosophy and Cognitive Science from Indiana University in 1999. From then to 2012, he taught at Franklin & Marshall College (F&M), where he was Professor of Psychology. In 2012, he became Professor of Philosophy and Psychology at the University of Cincinnati. Tony's research is both philosophical and empirical. It is focused on questions related to dynamical modeling, ecological psychology, artificial life and complex systems. He is author of more than 70 articles and the book *Radical Embodied Cognitive Science* (2009, MIT Press), which was a finalist for the Lakatos Award. His second book, co-authored with Stephan Kaufer, appeared on Polity Press in 2014. He is currently editing the second edition of the MIT Encyclopedia of the Cognitive Sciences.

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